

Balraj Ashwath

Machine Learning Engineer |  LinkedIn |  Google Scholar

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EDUCATION

- **National Institute of Technology Karnataka** Surathkal, IN
B.Tech in Electronics & Communications Engineering | GPA: 7/10 | Senior Thesis: 10/10 Aug. 2015 - Apr. 2019
- **Narayana Pre-University College** Bengaluru, IN
Karnataka State Pre-University | Percentage: 90% | JEE Mains: 99th %ile Jul. 2013 - May. 2015

EXPERIENCE

- **Microland Limited - Automation Team** Bengaluru, IN
Senior Machine Learning Engineer [Assoc Tech Arch - SW Dev] Jul. 2019 - Present
 - **Metric Time Series Analysis & Modeling:** Developed **Forecasting & Anomaly Detection** systems on metric data collected on edge devices. For *forecasting*, models such as *ARIMA*, *Prophet*, *NeuralProphet*, *NBEATS* & *TCN* were trained with ensemble MAPE at **6%**. AutoEncoders were used to perform Anomaly Detection with reconstruction *F1* at **81%**. High precision of forecasted-anomalies aided effective monitoring of edge environments.
 - **Alert Clustering:** Developed NLP-based **text clustering** system aimed at grouping unlabeled alerts into meaningful topics. Trained models such as *K-Means*, *LDA (TFIDF)*, *doc2vec*, *BERTopic* & other transformer-based models. *Tagging pipeline* was built to allow Ops Engineers to tag clusters and *thereby tagging any incoming alert to its closest cluster*. *Post-processing* methods including cluster correlation, new alert-type identification were applied to accurately tag more than **96%** of incoming alerts thereby *eliminating the need for a labeled alert corpus*.
 - Numerous modeling **POCs** involving Named Entity Recognition, Alert Classification, Alert Summarization, Hard Disk Defect Classification were conducted. *EDA & FE* were performed on various in-house & client datasets.
- **Video Analytics Lab, CDS Department, Indian Institute of Science** Bengaluru, IN
Research Intern (Mentors: Dr. R. Venkatesh Babu & K. Ram Prabhakar) May. 2019 - Jul 2019
 - **HDR Image Deghosting:** Developed novel CNN model for *motion compensation* in High Dynamic Range image-fusion & -deghosting. The proposed model improved SOTA on Khalantari SIGGRAPH 2017 dataset by **1.2 dB** PSNR (Peak Signal-to-Noise Ratio) score with **60%** faster inference times. [Work Presented at ECCV 2020]
- **Image Processing & Analysis Lab, National Institute of Technology Karnataka** Surathkal, IN
Undergraduate Researcher (Mentors: Dr. Shyam Lal & Karuna K. Eerapu) Jul. 2018 - Apr 2019
 - **Road Extraction from Satellite Imagery:** Developed novel segmentation architecture to identify road pixels in aerial imagery based on concepts from U-Net, dense & residual networks. The proposed model improved SOTA on Massachusetts Roads dataset by **2%** with an *order-of-magnitude lesser* parameters. [IEEE Access Journal Paper]
- **RBL Bank** Mumbai, IN
Summer Intern (Machine Learning) May 2018 - Jul. 2018
 - **e-KYC Process Automation:** Developed lightweight CNN-based OCR models to identify KYC details from PAN/ Aadhaar cards. Achieved **83%** F1 score with **sub-30ms** inference times on low-compute edge devices.

PUBLICATIONS

- **Dense Refinement Residual Network for Road Extraction from Aerial Imagery Data**, Karuna K. Eerapu*, Balraj Ashwath*, Shyam Lal, Fabio D. Acqua & A.V. Narasimhadhan, *IEEE Access Journal*, 2019. [* - equal contbⁿ]
- **Towards Practical and Efficient High-Resolution HDR Deghosting with CNN**, K. Ram Prabhakar, Susmit Agrawal, Durgesh Singh, Balraj Ashwath, & R. Venkatesh Babu, *European Conference on Computer Vision*, 2020.

SKILLS

ML Frameworks: PyTorch, TensorFlow, Keras | **Languages:** Python, Java, C, C++, R, Matlab, SQL, $\LaTeX$

Libraries: Pandas, Scikit-learn, NumPy, Plotly, OpenCV, Darts | **Tools:** Git, MongoDB, Elasticsearch, MySQL, Tableau

ACHIEVEMENTS / CERTIFICATIONS / ACTIVITIES

- Shortlisted in Microsoft's Incubathon 2021 | Silver Medalist in Kaggle's NFL Impact Prediction 2021 - Top 10%
- **Certifications:** Azure AI Engineer Associate (AI-100) | **Coursera:** Deep Learning Specialization (deeplearning.ai)
- Reading latest research papers & blog posts in AI / ML & experimenting models | Participating in Kaggle competitions.